

4. Planning Project

Q.1 Explain the crashing of the project. Explain with a small example the process of crashing.

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- Project crashing is when you shorten the duration of a project by reducing the time of one or more tasks.
- crashing is done by increasing the resources to the project, which helps make tasks take less time than what they were planned of.
- Of course, this also adds to the cost of the overall project.
- Therefore, the primary objective of project crashing is to shorten the project while also keeping costs at a minimum.
- There are several key players involved in project crashing, and it's a collaborative process.

1) Project Managers

- ⇒ Project managers are the central figures as they are responsible for deciding when and how to crash a project.
- They help manage resources and oversee changes made to the project schedule.
- The project manager will communicate any changes to all stakeholders.

2) Team members

- ⇒ Team members are also involved as they might need to work longer hours or

- They must cooperate and remain flexible.

3) Major stakeholder

⇒ Stakeholders such as clients or sponsors may drive the need for project crashing due to changing requirements or deadlines.

→ Stages:

1) Find critical Path

⇒ Find the normal critical path and identify the critical activity.

2) Calculate Cost Slope

⇒ Calculate the cost slope for the different activities by using the formula:

$$\text{Cost slope} = \frac{\text{Crash cost} - \text{Normal cost}}{\text{Crash time} - \text{Normal time}}$$

The cost slope indicates the extra cost required to expedite an activity per unit time.

3) Ranking

⇒ Rank the activities in the ascending order of cost slope.

- The activity having the minimum cost slope have to be crashed. First, crash the selected activity to its minimum duration.

4) Crashing

- ⇒ Crash the activities in the critical path as per the ranking i.e., activity having lower cost slope would be crashed first to the maximum extent possible.
- Calculate the new direct cost by cumulative adding the cost of crashing to the normal cost.

5) Parallel Crashing

- ⇒ As the critical path duration is reduced by the crashing in step 3, other paths also become critical, i.e., we get parallel critical path.
- This means that project duration can be reduced duly by simultaneous crashing of activity on the parallel critical path.

6) Total cost

- ⇒ Crashing as per step 3 and 4, one reaches a point when further crashing is either not possible or does not result in the reduction of crashing or project duration.
- For the different project durations total cost is found up to total cost is got by adding corresponding fixed cost to the direct cost, and the direct cost is got by adding the expediting crashing cost cumulative to the normal cost.

- For e.g., Project Crashing in IT.
- IT projects can also benefit from project crashing, and the focus is typically on shortening the critical path.
- However, this can lead to challenges like rushed development, integration issues, security risks and more.
- For example, if a company is working on upgrading its operating system by a particular deadline, it may bring in outside contractors and consultants to help while the internal team works extended hours.
- This increases costs but keeps the project on schedule.

Q.2 What is differences between resource loading and resource levelling.

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1) Resource Loading

⇒ Resource loading involves assigning resources to tasks in a project schedule.

- It helps in identifying the total quantity of resources required for each task and the time at which they are needed.

- For example,

If you are managing a construction project, you might use resource loading to determine how many workers, machines and materials are needed each day.

2) Resource Leveling

⇒ Resource leveling, on the other hand, is a technique used to smooth out the resource allocation when resources have been over-allocated or under-allocated.

- It aims to minimize the fluctuation of resource usage and ensure a consistent workflow.

- For e.g.,

In the same construction project, if you find that you have too many workers scheduled for one day and not enough for another, you would use resource leveling to redistribute the workers more evenly across the project timeline.

Resource Loading

1) Assigns resources to tasks.

2) Identifies total resources needed.

3) Used in the planning phase.

4) Ignores project time constraints; purely about loading data.

Resource Leveling

1) Redistribute resources.

2) Minimizes resource usage fluctuation.

3) Used in the execution phase.

4) Balances resource availability while respecting constraints.

Q.3 Explain Goldratt's 'Critical Chain'

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- Critical chain is a schedule network analysis technique that takes into account task dependencies, limited resource availability and buffers necessary to successfully complete the project.
- CCM allows a project manager to plan and execute project's schedule by concentrating on all resources used in critical path.
- This is a critical technique intended to speed up the process by improving the rate of meeting due dates.
- It focuses on planning and managing by paying attention to resources and resource management, risks, uncertainties.
- Steps:

1) Identify the critical chain

⇒ The first step in the critical chain project management is to identify the critical chain, that runs alongside sequentially dependent tasks, considering the availability of resources as constraints.

- The considerations start with a deep investigation of the schedule network diagram which illustrates different duties and their inter-dependencies.
- Resource limitations are identified next

2) Schedule and Buffer management

⇒ The critical chain has been identified and we will build a project schedule around it, doing our best to allocate resources to critical tasks to avoid them running late.

- By buffers, uncertainties and variability are factored in to enable proper task flow with feeding buffers being placed before non-critical jobs.

3) Resource allocation and Management

⇒ Resource allocation and management being often portions of CPM, space-holders include decision-making on the allocation of resources to important job activities in the critical chain.

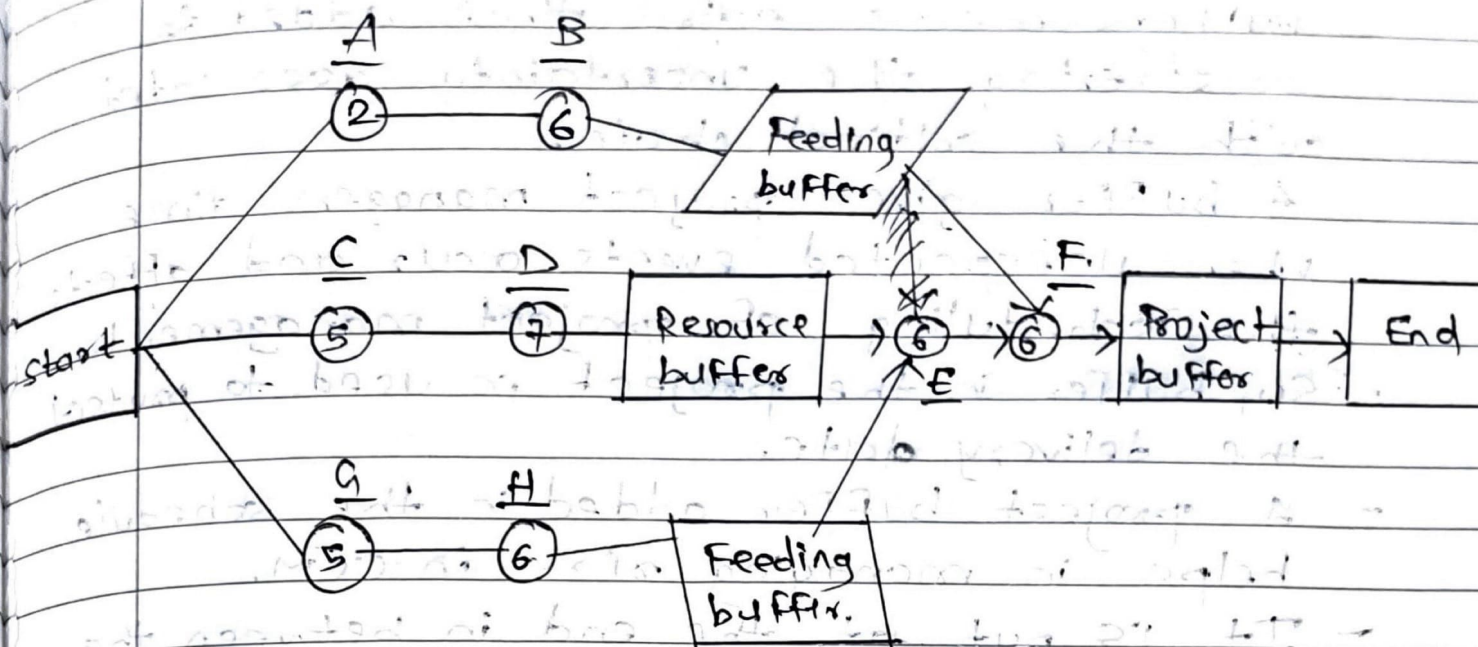
- Resource buffers are used effectively to protect key tasks from delays

4) Buffer Monitoring and Control

⇒ Additionally, buffer monitoring and control provide accurate and detailed project reporting and timely identification of issues that can affect project development.

- Ongoing assessment of buffer use permits the manager to realize the risks and uncertainties ahead of situation

- Critical Chain Network Diagram:



Path: Start → A → C → D → E → F → End

- (1) Project buffer
- (2) Feeding buffer
- (3) Resource buffer
- (4) Capacity buffer

Project buffer is the last buffer in the project network. It is used to protect the project completion date from delays in the non-critical chain.

Q.4 Project Buffer

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- Buffers means extra time added considering the uncertainty associated with the critical chain.
- A buffer gives project managers time when unexpected events occur and affects the scheduling of project management.
- So, buffer in the project is used to protect the delivery date.
- A project buffer added in the schedule helps in managing risk in CCM.
- It is put at the end in between the last task and the date of completion.
- It helps to keep the completion date the same despite any delays.
- Those delays consume buffer time which allows protecting the completion date.
- There are four types of buffers!

- 1) Project Buffer
- 2) Feeding Buffer
- 3) Resource Buffer
- 4) Capacity Buffer

1) Project Buffer

⇒ Protects the project from missing its scheduled end date and keeps the completion date unchanged.

- It is inserted at the end of the project network diagram, between the last task and the completion date.
- It protects project completion date, which might vary due to the changes in activity durations in the critical chains.

2) Feeding Buffer

- ⇒ It is inserted between the last task on a non-critical chain and critical chain.
- These buffers are typically added to a non-critical chain don't affect the critical chain.

3) Resource Buffer

- ⇒ These are added to the critical chain to ensure appropriate resources are available throughout the projects when needed.
- These resources are commonly known as Critical Resources.

4) Capacity Buffer

- ⇒ It is added on call resources necessary in case unforeseen budget issue arise.

Q.5 Explain the risk breakdown structure.

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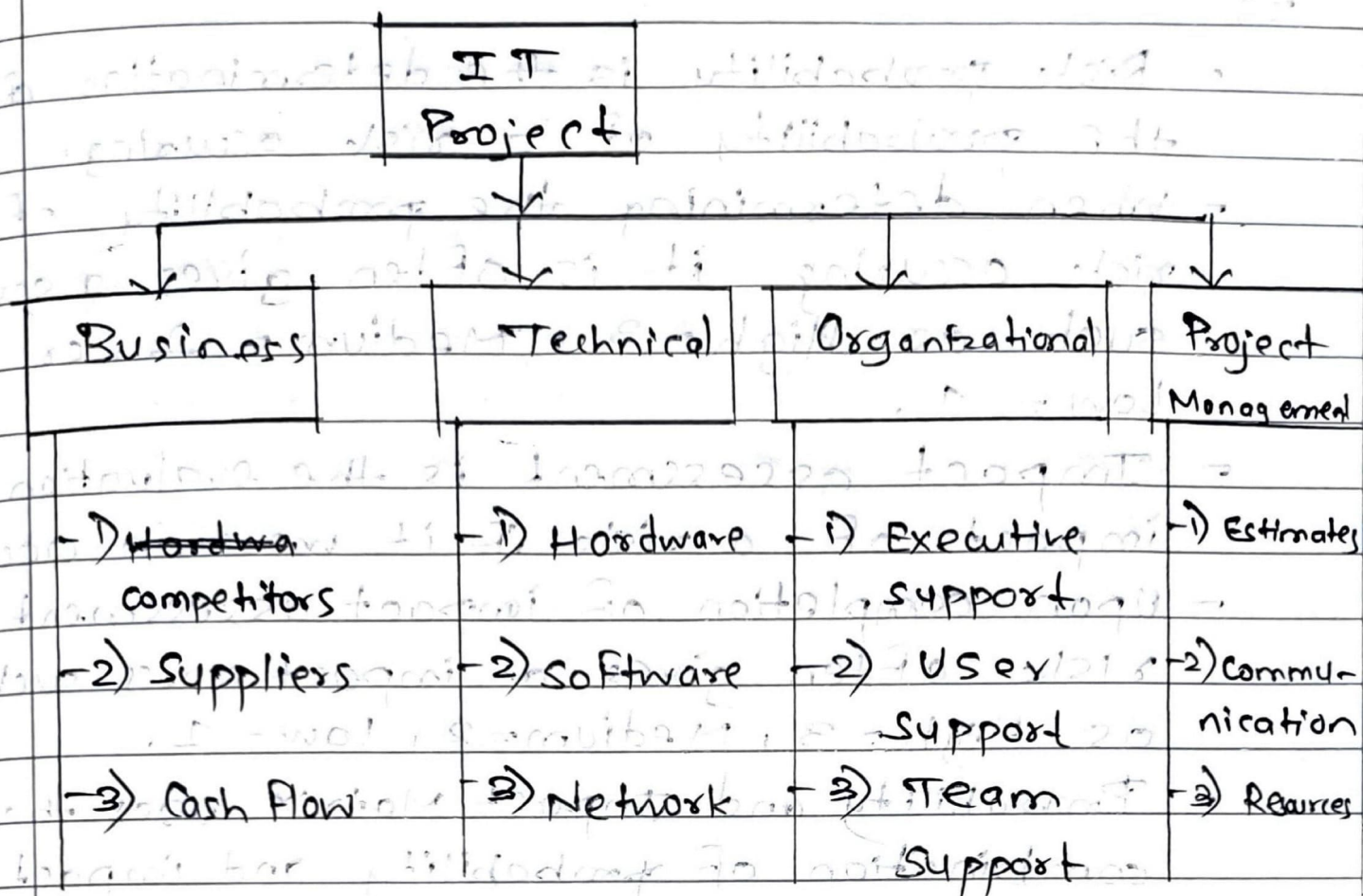
- A risk breakdown structure is a hierarchy of potential risk categories for a project.
- A risk breakdown structure is a tool for managing risks, which are events that you ~~not~~ have not planned for or expected.
- Similar to WBS, but used to identify and categorize risk.
- However, there are also positive risks that can benefit a project.
- Either way, project managers have to prepare for risks, either good or bad it can interfere with project objectives.
- A risk breakdown structure breaks down risks in hierarchical graph, beginning at the higher level and moving down to the finer level risks.
- RBS structure follows a tree-like structure, breaking risks into multiple levels:

i) Level 1: Broad risk categories

ii) Level 2: Subcategories within each category.

iii) Level 3 & Below: Specific risks under each subcategory.

For e.g.,



1) Business Risk : Competitor releases a similar product early, threatening market share.

2) Technical Risk : A critical software module fails under peak load, causing service downtime.

3) Organizational Risk : Lack of executive sponsorship, delays decision-making and resource approval.

4) PM Risks : Underestimating task durations leads to schedule overruns and budget strain.

Q. 6 Explain probability and impact matrix.

⇒

- Risk probability is the determination of the probability of a risk occurring.
- When determining the probability of a risk occurring it is often gives a score such as High = 3, Medium = 2, or Low = 1.
- Impact assessment is the evaluation of impact of a risk if it were to occur.
- Upon completion of impact assessment a risk often gives an impact scores such as High = 3, Medium = 2, Low = 1.
- Probability and Impact Matrix uses the combination of probability and impact scores of individual risks and ranks or prioritizes them for easy handling of the risks.

Preparation of Probability & Impact Matrix

1) Determine data to use

⇒ While preparing the matrix, the first thing to be done is to decide on the type of data that will go into your matrix.

2) Decide on the size of your matrix

⇒ Next is to decide on the size of your matrix.

For eg., 3x3, 5x5

3) List the events to include in risk matrix

⇒ Next is to list all events to be entered into the risk matrix.

4) Gather Impact and Probability Data

⇒ After listing the events, gather impact and probability data for each event.

5) Enter Impact and Probability Data

⇒ Enter final data for each event into the Event coordinate table.

6) Determine how to categorize impact data.

7) Determine how to categorize probability data.

- For eg.,

		Impact				
		Trivial	Minor	Moderate	Major	Extreme
Probability	Rare	Low	Low	Low	Medium	Medium
	Unlikely	Low	Low	Medium	Medium	Medium
	Moderate	Low	Medium	Medium	Medium	High
	Likely	Medium	Medium	Medium	High	High
	Very likely	Medium	Medium	High	High	High

Q.7 Explain risk response strategies for positive and negative risks.

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- A risk is not always negative.
- A risk is known as a positive risk or opportunity if the impact is positive which you may want to actualize.
- Similarly, a risk is known as negative risk or threat if the impact is negative, and in this case, you will want to lessen its impact.

• Negative Risks

1) Escalate

⇒ You can use the escalate risk response strategy when you identify risk and find that you cannot manage it on your own because you lack the authority, resources or knowledge required for a response.

- For example, you know that the government is planning to announce a regulation and it could impact your project negatively if approved.
- You have no legal advisor and other resources to manage the risk so you will approach your supervisors to handle the risk.

2) Mitigate

⇒ You will try to lessen the impact of risk in this risk response strategy.

- You can do so by either trying to decrease the probability of the risk happening or the effect of the risk.
- For e.g., you find that a team member may leave for a specific duration during the peak of your project.

3) Transfer

- ⇒ You use this strategy when you lack skills or resources to manage the risk, or you are too busy to manage it.
- In the transfer risk response strategy, you transfer the risk to a third party to manage it.
- For e.g., in your project, there is a task to install some requirement equipment and you ~~find~~ have little experience with this task.
- The task is complex, and few contractors have successfully installed it.
- Therefore, you find a contractor and ask them to install it.

4) Avoid

- ⇒ Here, you try to eliminate the risk or its impact.
- You do this by changing your project management plan, by changing the project scope, or by changing the schedule.
- For e.g., Using a different supplier to avoid delivery delays.

Positive Risk Response Strategy

1) Enhance

⇒ In enhance risk response strategy you try to increase the chance of risk happening so you can realize the benefits of the risk.

→ In this case, you try to realize the opportunity.

→ You can say that enhance risk response strategy is the opposite of mitigation risk response strategy.

→ For e.g., OFFERING incentives to suppliers for early delivery.

2) Exploit

⇒ In the exploit risk response strategy, you ensure that the opportunity is realized.

→ Here, you do not try to realized the opportunity; Here you make sure that the opportunity realized.

→ For example, Assigning best team members to fast-track task to gain early competition hours.

3) Accept

⇒ This risk response strategy is common for both types of risks, i.e. positive and negative risks.

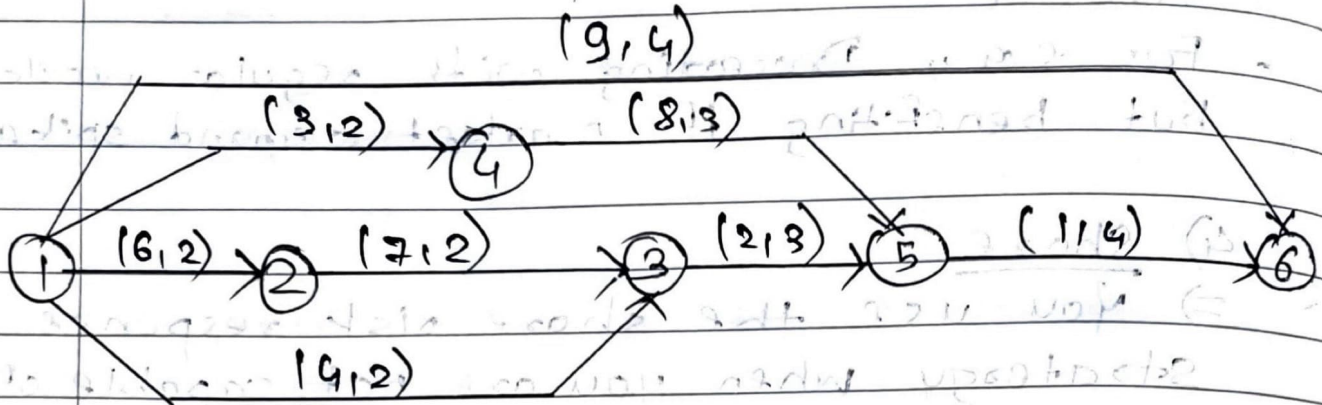
→ In the accept risk response strategy you take no action to realize the opportunity.

- You leave the opportunity as is and if it happens on its own, you will benefit from it
- For e.g., Proceeding with regular work but benefiting if market demand spikes.

4) Share

- ⇒ You use the share risk response strategy when you are not capable of realizing the opportunity on your own.
- So you team up with another company and work together to realize the opportunity.
- For e.g., Partnering with another company to co-develop a high-demand product.

Q.8 (May 22) (9.3 E)
 ⇒ Solⁿ :-

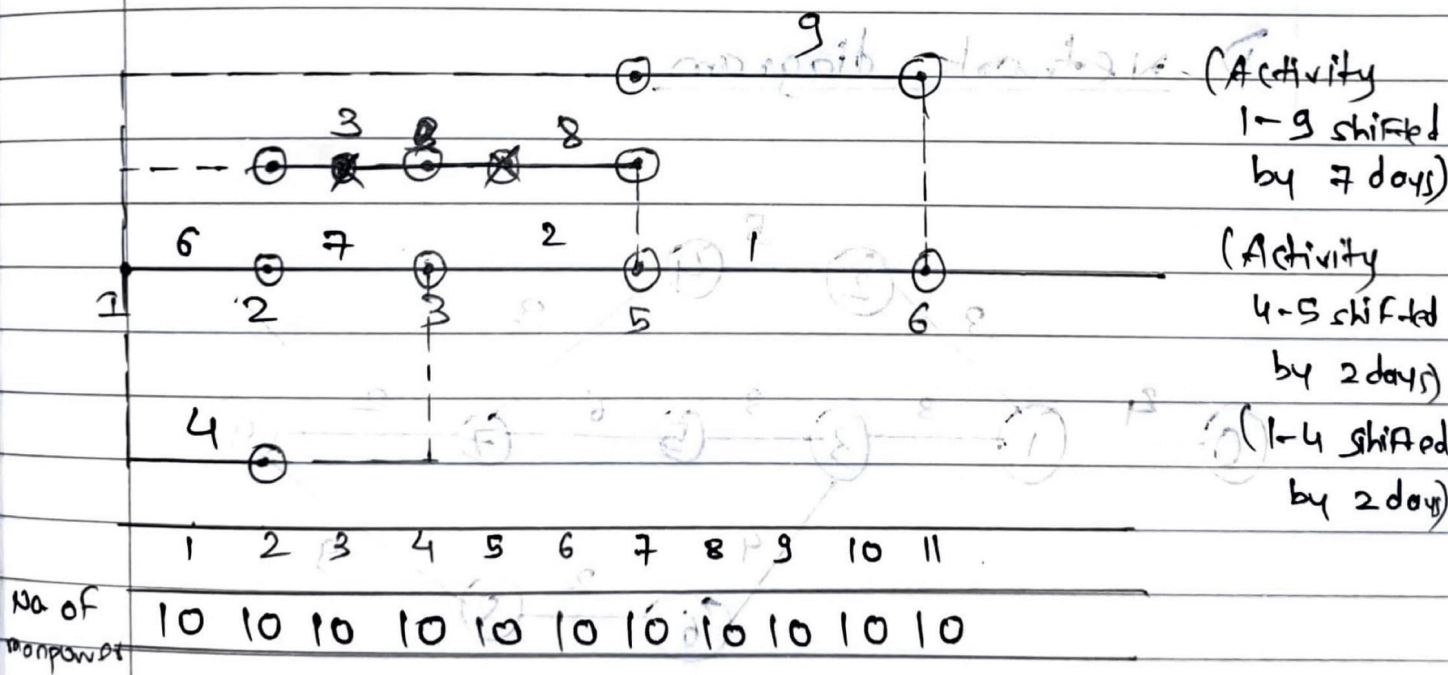
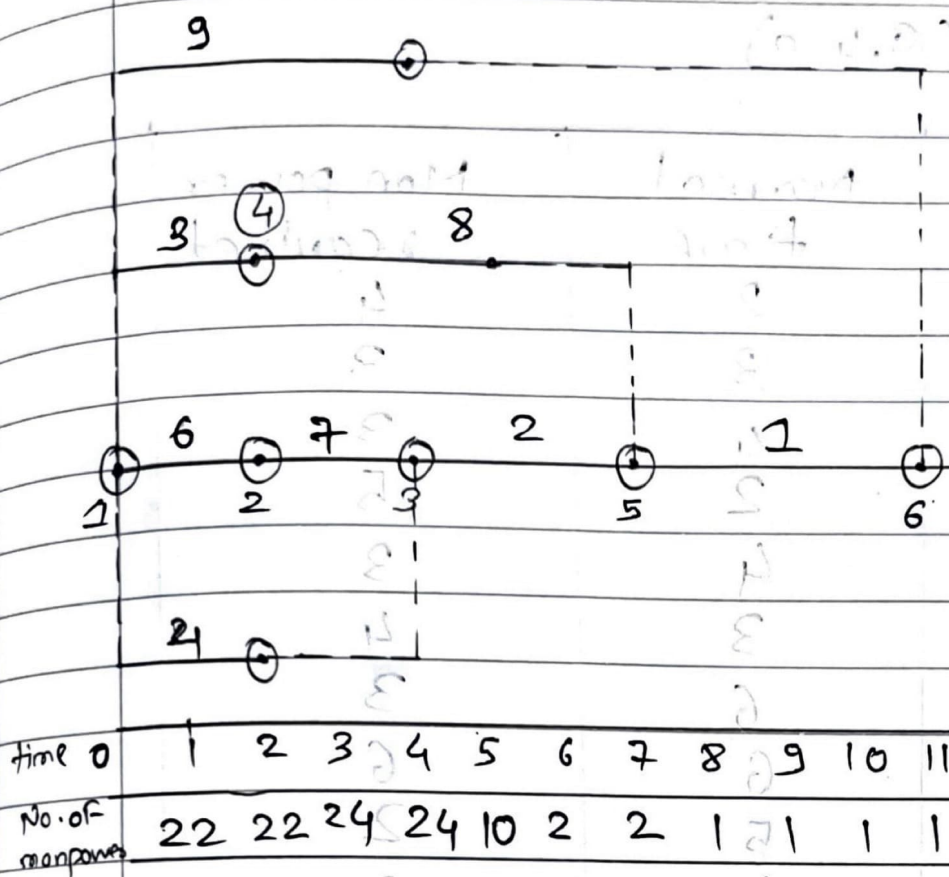


Activity	time	Manpower
1-2	2	6
1-3	2	4
1-4	2	3
1-6	4	9
2-3	2	7
3-5	3	2
4-5	3	8
5-6	4	1

- Paths :

- 1-2-3-5-6 : $2+2+3+4 = 11$ days
- 1-4-5-6 : $2+3+4 = 9$ days
- 1-3-5-6 : $2+3+4 = 9$ days
- 1-6 : 4 days

So, the critical path is :
 1-2-3-5-6



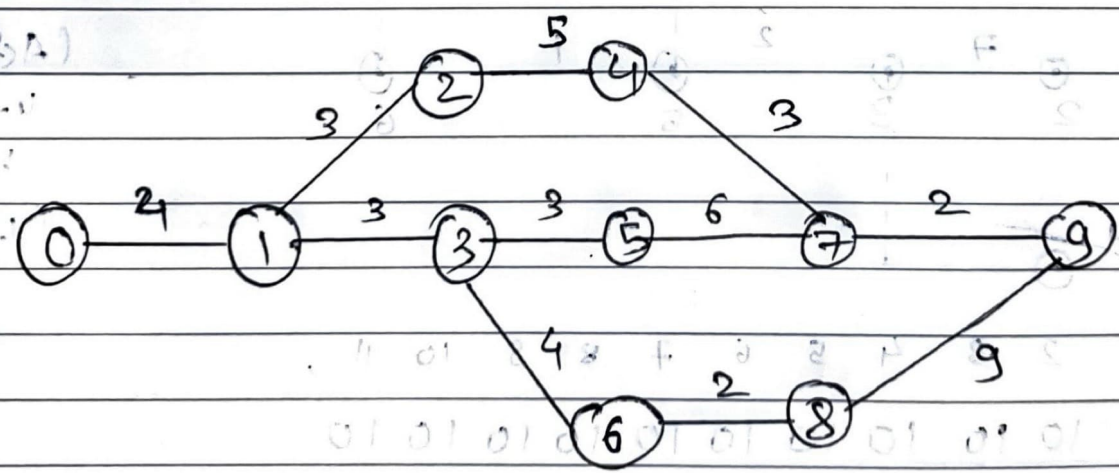
No. of men required to complete project in 11 days = 10 days

Q.9 (Dec 23) (Q.4 a)

=>

Activity	Normal time	Man power required
0-1	2	4
1-2	3	3
1-3	4	3
2-4	2	5
3-5	4	3
3-6	3	4
4-7	6	3
5-7	6	6
6-8	5	2
7-9	4	2
8-9	4	9

Network diagram



Paths:

a) 0 - 1 - 3 - 5 - 7 - 9

=> 2 + 4 + 4 + 6 + 4 = 20 ✓

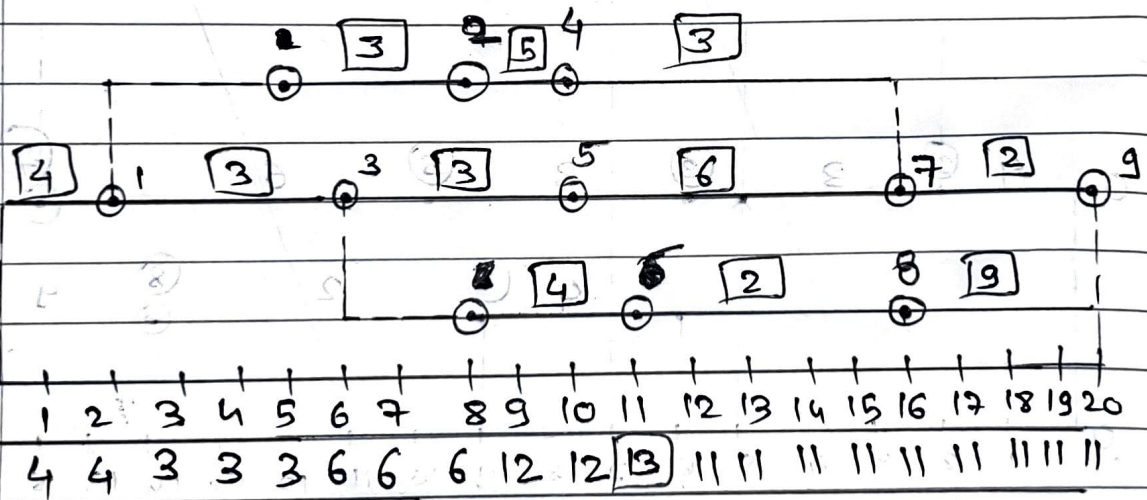
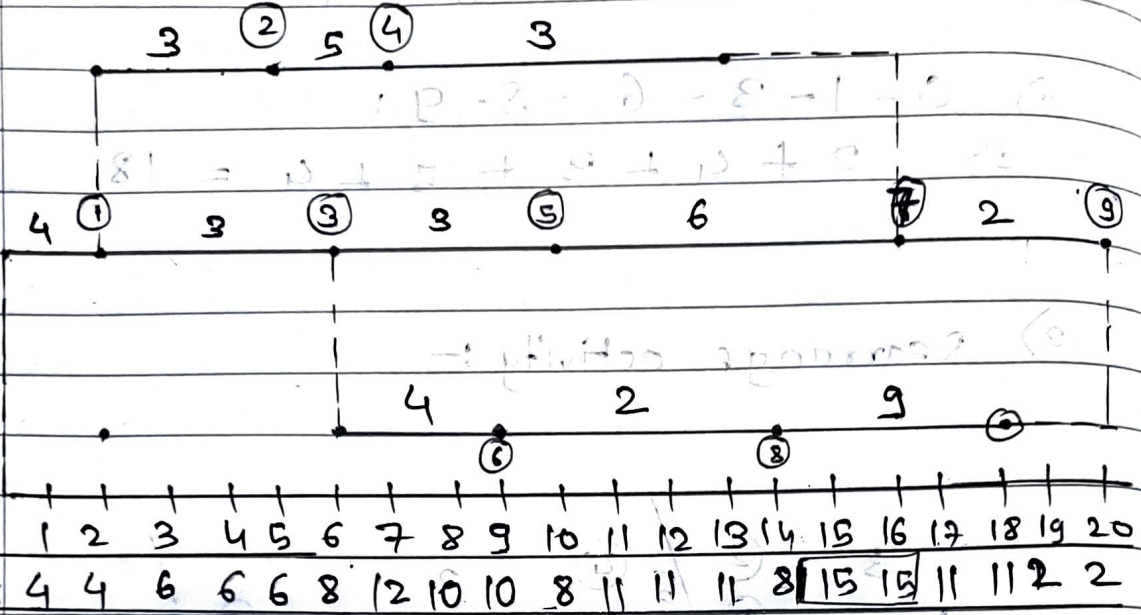
$$b) \quad 0-1-2-4-7-9:$$

$$\Rightarrow 2 + 3 + 2 + 6 + 4 = 17$$

$$c) \quad 0-1-3-6-8-9:$$

$$\Rightarrow 2 + 4 + 3 + 5 + 4 = 18$$

2) Rearrange activities



Q.14 What is risk management? Construct a risk table for five risk identified in your project.

⇒

- Risk management is a process of identifying, analysing and responding to project risks to minimize negative impact and maximize positive impacts.
- It ensures project success by systematically managing uncertainties.
- Risk management process consists of:
 - 1) Risk Identification
 - 2) Risk Assessment / Analysis
 - 3) Risk Planning
 - 4) Risk Monitoring

1) Risk Identification

⇒ Risk identification refers to the systematic process of recognizing and evaluating potential threats or hazards that could negatively impact an organization, its operation.

2) Risk Analysis

⇒ Risk analysis is the process of evaluating and understanding the potential impact and likelihood of identified risk on the organization.

- It helps determine how serious a risk is and how to best manage or mitigate it.

3) Risk Planning

⇒ Risk planning involves developing strategies and actions to manage and mitigate identified risks effectively.

4) Risk Monitoring

⇒ Risk monitoring involves continuously tracking and overseeing identified risks to assess their status, changes and effectiveness of mitigation strategies.

- For e.g., E-commerce Website Development

	Risk description	Probability	Impact	Response strategy
1)	Website performance issue during peak traffic	High	High	Mitigation
2)	Security vulnerabilities leading to data breaches	High	High	Avoid
3)	Payment gateway integration failure	Medium	High	Transfer
4)	Delay in product catalog updates	High	Medium	Mitigation
5)	Scope creep due to frequent client change request	High	Medium	Accept

9 Explain Project Communication Management.

- In project management, effective communication is vital for smooth information flow among involved parties.
- Good communication is crucial because miscommunication can lead to delays, cost overruns and project failure.
- Creating Your Project Communication Plan:

Step 1: Navigating Communication Objectives



Step 2: Project stakeholders overview



Step 3: Project Update



Step 4: Choosing right communication channel



Step 5: Timeline and Reporting

- 1) Navigating Communication Objectives
- ⇒ Establishing clear communication goals is the first crucial step in crafting an effective Project Communication Management Plan.
- Define your objectives, whether it's creating awareness through status reports or seeking approval from sponsors.

2) Project Stakeholder Overview

- ⇒ Identify all individuals involved in project, including team members, sponsors, customers and other stakeholders.
- Consider those who may be impacted or have influence for a comprehensive list.

3) Project Update

- ⇒ Provide friendly and engaging project update message to stakeholders.
- Communication that the project is on track regarding scope, schedule, budget, objectives, risks and deliverables.

4) Choosing the right communication channels

- ⇒ Consider the audience and context when sharing project information.
- Opt for formal communication channels like email reports for stakeholders or informal verbal debriefs during team meetings.

5) Timeline and Reporting

- ⇒ Establish clear timeline for communication.
- Determine whether stakeholders require weekly, or monthly reports and identify any specific deadlines.
- Consider different time zones and varied employee schedule to accommodate everyone.